AMYLOID DATA AGGREGATOR AND CURATOR WEB APP

A PREPRINT

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October 23, 2026

ABSTRACT

I present a web-based resource that pools deposited amyloid structures of proteins involved in human diseases in order to facilitate a residue-focused study. Amyloid Data Aggregator and curator (ADA) combines deposited PDB structural data with secondary structural annotations for a residue-centric view of amyloid structures. The starting point for ADA is the Amyloid Atlas - a web-based resource for amyloid PDB structures. Secondary structural annotations are assigned according to the corresponding mmCIFs in the RCSB Protein Data Bank (PDB). A central goal of this web app is to provide researchers with a community-maintained resource. The data is programmatically aggregated from established resources including the aforementioned sites, but is open to manually-entered corrections that can be submitted for any row of the PDB table, authorised via an ORCID sign-in. These submitted changes would then be manually accepted or rejected by myself (and ideally other users assigned 'maintainer' status in future). The full history of submitted changes and their outcomes is made visible (similar to Wikipedia).

Keywords amyloid structure · bioinformatics

1 Introduction

Substantive advances in methods for structure determination has led to near-atomic resolution of disease-associated amyloid fibrils. Such information had previously only been possible for crystalline forms of amyloid made of short peptides ref and X-ray fibre diffraction of dessicated fibres ref. Solid state Nuclear Magnetic Resonance (ssNMR) spectroscopy had also provided varyingly accurate structures ref. Advances in electron cryo-microscopy led primarily by .. in 2014 ref and by Bayesian approaches to solving the structure ref Scheres led the way to published structures. The most striking and disease-relevant of these followed developments in tissue extraction methods, specifically paired helical filaments and straight filaments found in the brains of patients diagnosed with Alzheimer's disease, at post-mortem ref. The number of deposited amyloid structures has steadily grown making the usefulness of a single repository for these apparent. This was addressed by Amyloid Atlas, built and maintained by the Sawaya and Eisenberg labs at UCLA ref. The resource includes physicochemical representations of a single chain of each PDB. A similar research resource that reads from Amyloid Atlas provides a series of other computational analyses made possible by the deposited coordinates of each protein structure

The original purpose for ADA was for personal use, as a way to compare outputs of an algorithm (called SALSA) with solved PDB structures, specifically the location of the amyloid core and beta-sheets therein. It grew out of this very simple requirement as I steadily added more and more features. Gradually the design's guiding principle went from 'what functionality do I want' to 'what functionalities might others find useful too'.

These manually curated changes persist, not to be overwritten by the periodic automated reloads of data from the different sources like Amyloid Atlas and the RCSB PDB.

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

2 Implementation details

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2.1 Application design

The design of the application was primarily for an Internet-based web application to be run entirely in the browser. It is fully written in Typescript with React.js, HTML and CSS. It is deployed on web-hosted platform Render (free-tier). The introduction of the manual curation feature provided a requirement for integration with authorisation via ORCID. The storage of edit histories created the requirement for a persistent data store. I opted for a web-hosted Postgres database, supplied by Neon (free-tier).

2.1.1 Headings: third level

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3 Examples of citations, figures, tables, references

3.1 Citations

Citations use `natbib`. The documentation may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Here is an example usage of the two main commands (`citet` and `citep`): Some people thought a thing [Kour and Saabne, 2014a, Hadash et al., 2018] but other people thought something else [Kour and Saabne, 2014b]. Many people have speculated that if we knew exactly why Kour and Saabne [2014b] thought this...

3.2 Figures

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[2] Sample of the first footnote.

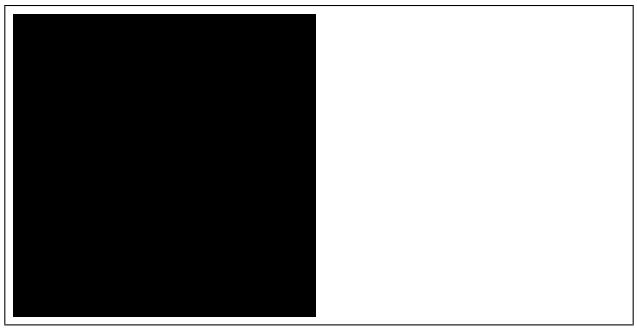


Figure 1: Sample figure caption.

Table 1: Sample table title

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

3.3 Tables

See awesome Table 1.

The documentation for `booktabs` ('Publication quality tables in LaTeX') is available from:

`https://www.ctan.org/pkg/booktabs`

3.4 Lists

- Lorem ipsum dolor sit amet
- consectetur adipiscing elit.
- Aliquam dignissim blandit est, in dictum tortor gravida eget. In ac rutrum magna.

References

George Kour and Raid Saabne. Real-time segmentation of on-line handwritten arabic script. In *Frontiers in Handwriting Recognition (ICFHR), 2014 14th International Conference on*, pages 417–422. IEEE, 2014a.

Guy Hadash, Einat Kermany, Boaz Carmeli, Ofer Lavi, George Kour, and Alon Jacovi. Estimate and replace: A novel approach to integrating deep neural networks with existing applications. *arXiv preprint arXiv:1804.09028*, 2018.

George Kour and Raid Saabne. Fast classification of handwritten on-line arabic characters. In *Soft Computing and Pattern Recognition (SoCPaR), 2014 6th International Conference of*, pages 312–318. IEEE, 2014b. doi:10.1109/SOCPAR.2014.7008025.